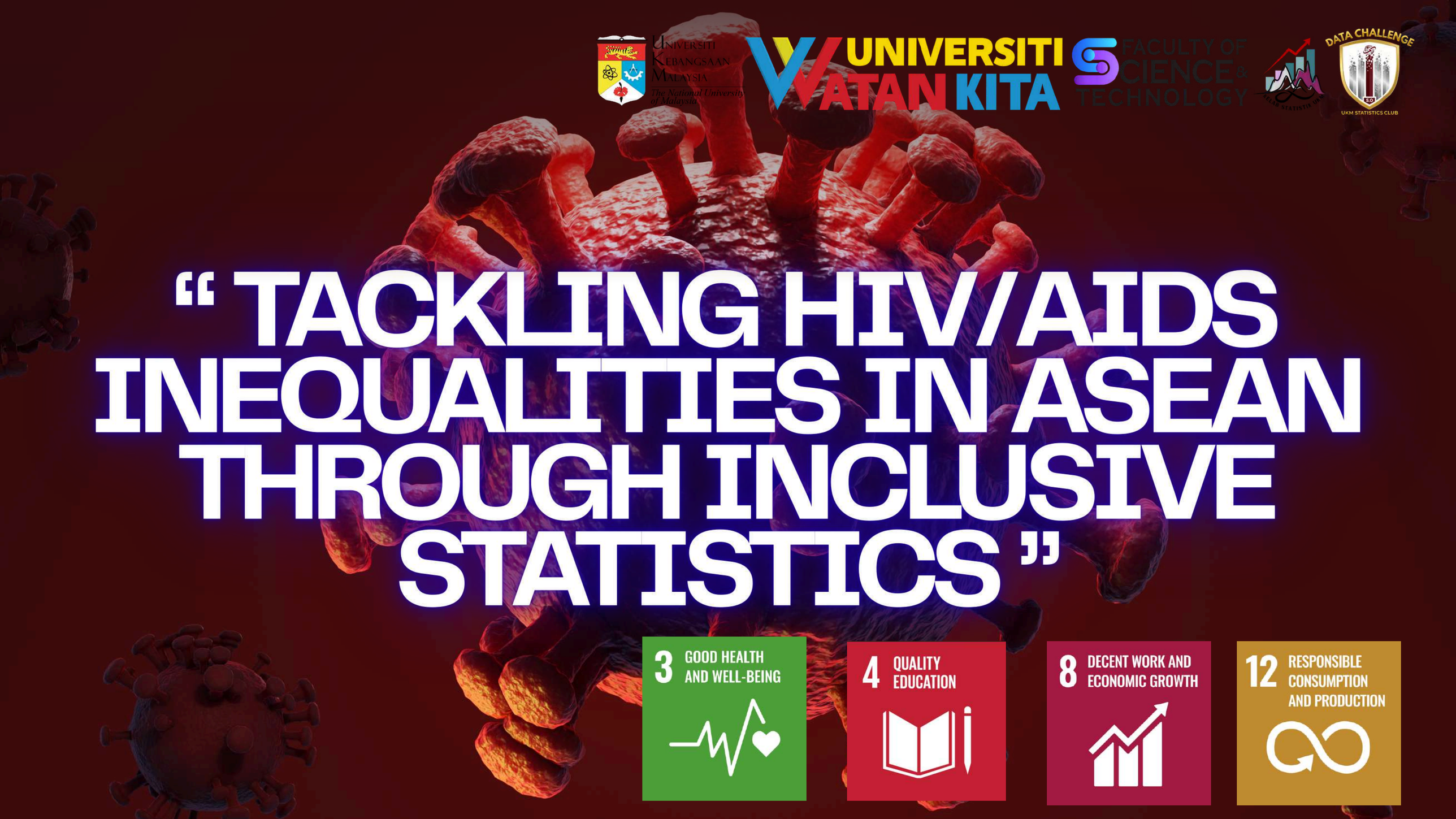

Universiti Kebangsaan Malaysia
The National University of Malaysia
UNIVERSITI VATAN KITA
FACULTY OF SCIENCE & TECHNOLOGY
DATA CHALLENGE
UKM STATISTICS CLUB
“ TACKLING HIV/AIDS INEQUALITIES IN ASEAN THROUGH INCLUSIVE STATISTICS ”
3 GOOD HEALTH AND WELL-BEING
4 QUALITY EDUCATION
8 DECENT WORK AND ECONOMIC GROWTH
12 RESPONSIBLE CONSUMPTION AND PRODUCTION

TEAM MEMBERS



DR WAN NURAINI FARHANA BINTI WAN NASIR



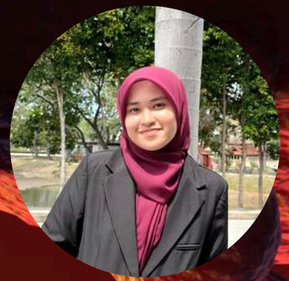
Naufal Akasyah bin Bakri



Mohamad Afiq Muttaqin Bin Muttacohiri



Nurul Dyaasykin Binti Mohd Ezany



Nur Ayuni Binti Azizi



Che Wan Nur Dayana Binti C Wan Roslan

INTRODUCTION

THE HUMAN IMMUNODEFICIENCY VIRUS (HIV) AND ACQUIRED IMMUNODEFICIENCY SYNDROME (AIDS) REMAIN SIGNIFICANT PUBLIC HEALTH CHALLENGES IN THE ASSOCIATION OF SOUTHEAST ASIAN NATIONS (ASEAN) REGION.

AIDS (Acquired Immunodeficiency Syndrome)

AIDS stands for Acquired Immunodeficiency Syndrome. It is the most advanced stage of HIV infection.

When HIV is not treated, it slowly damages the immune system until the body can no longer fight infections. At this stage, the person develops AIDS.

People with AIDS may develop opportunistic infections, which are diseases that attack when the immune system is very weak, such as:

- Tuberculosis (TB)
- Pneumonia
- Certain cancers
- Severe infections



SOURCE : THE STAR NEWS

HIV (Human Immunodeficiency Virus)

HIV stands for Human Immunodeficiency Virus. It is a virus that attacks the immune system, especially the CD4 cells (T-helper cells) that help the body fight infections.

Common ways HIV spreads:

- Unprotected sexual contact with an infected person
- Sharing needles or syringes
- From mother to baby during pregnancy, birth, or breastfeeding
- Blood transfusion with infected blood (rare today because blood is screened)

A person can live with HIV for many years if they receive proper treatment called antiretroviral therapy (ART).



OBJECTIVES OF THE STUDY CASE



ANALYZE

To examine the distribution of HIV/AIDS cases and treatment across ASEAN countries using statistical data.



DATA-DRIVEN STRATEGIES

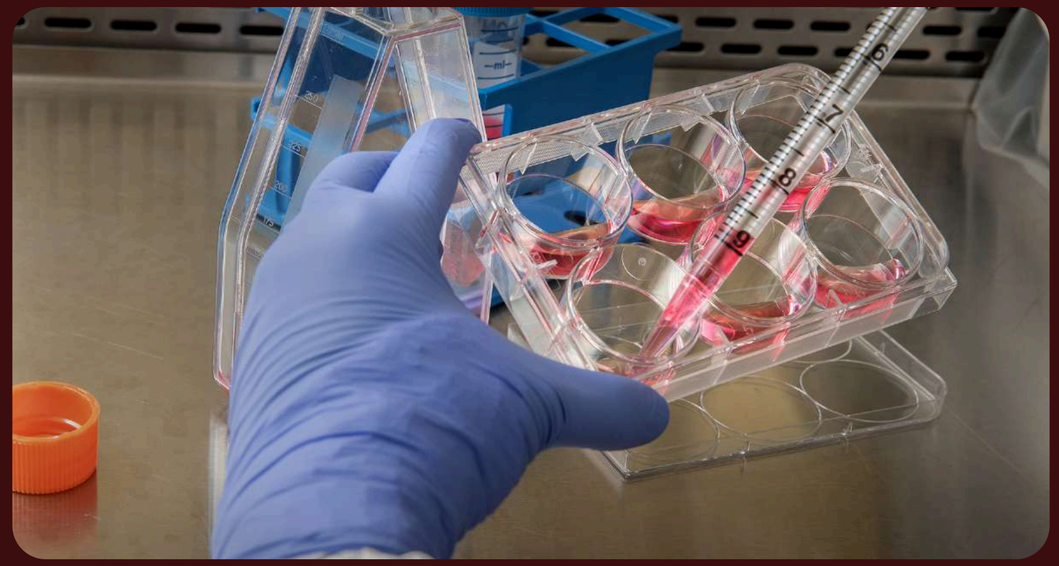
To propose data-driven recommendations to reduce HIV/AIDS inequalities in ASEAN based on inclusive statistical analysis.



EVALUATION

To evaluate how inclusive statistics can help highlight health disparities related to HIV/AIDS in the ASEAN region.

The project supports Sustainable Development Goal 3: Good Health and Well-being by using inclusive statistics to identify and reduce HIV/AIDS health inequalities in ASEAN, and contributes to Sustainable Development Goal 4: Quality Education by promoting awareness, knowledge, and evidence-based learning about HIV/AIDS through data analysis.



METHODOLOGY

STEP 1

Data Collection



STEP 2

Data Cleaning



STEP 3

Exploratory Data Analysis (EDA)



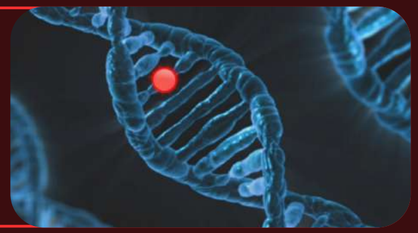
STEP 4

Statistical Analysis



STEP 5

Inequality Analysis



STEP 1 — DATA COLLECTION

The dataset was collected from trusted international health organizations such as:



World Health Organization



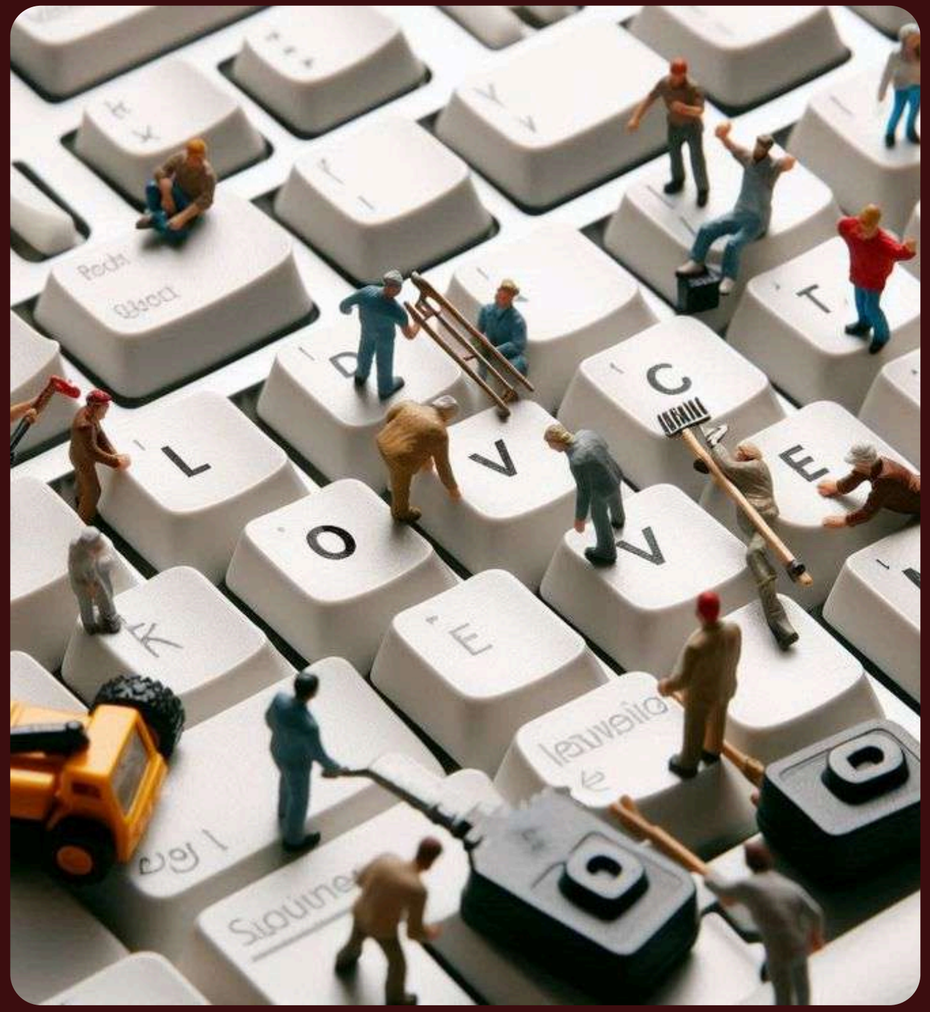
World Bank



UNAIDS

STEP 2 — DATA CLEANING

Data preprocessing was performed using Python to ensure data quality and consistency.



The cleaning process included:

- ***converting variables from text format to numeric format***
- ***replacing "No data" values with missing values***
- ***extracting numerical estimates from range values***
- ***handling missing values using median imputation***
- ***verifying data types and removing inconsistencies***

The cleaned dataset was then used for further analysis and visualization.

These steps ensure the reliability and accuracy of the analysis results.

STEP 3 — EXPLORATORY DATA ANALYSIS (EDA)



Exploratory data analysis was conducted using Microsoft Power BI to understand the distribution of HIV burden and ART coverage across countries.

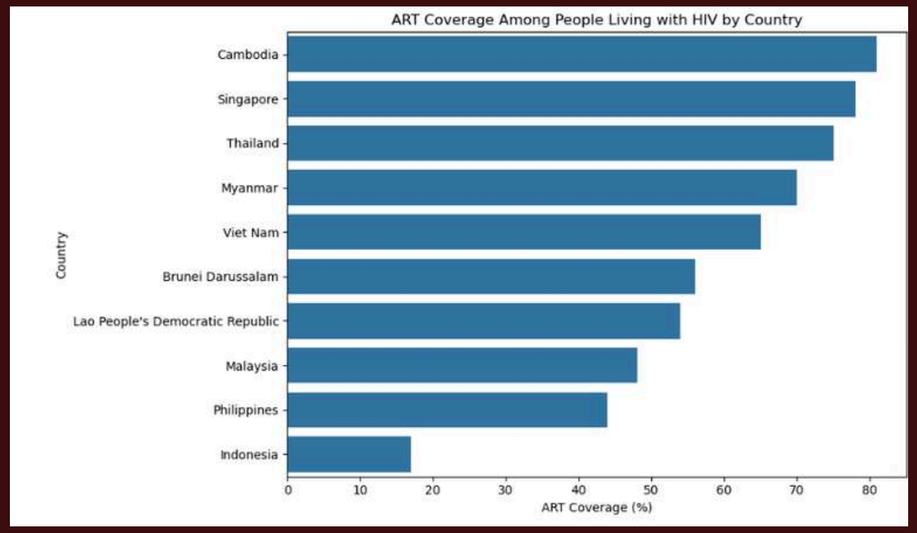
Key visualizations include:

- ***HIV burden by country (estimated number of people living with HIV)***
- ***ART coverage by country***
- ***Comparison of ART coverage and HIV burden across ASEAN countries***

This step helps identify patterns, trends, and disparities in HIV/AIDS distribution within the region.

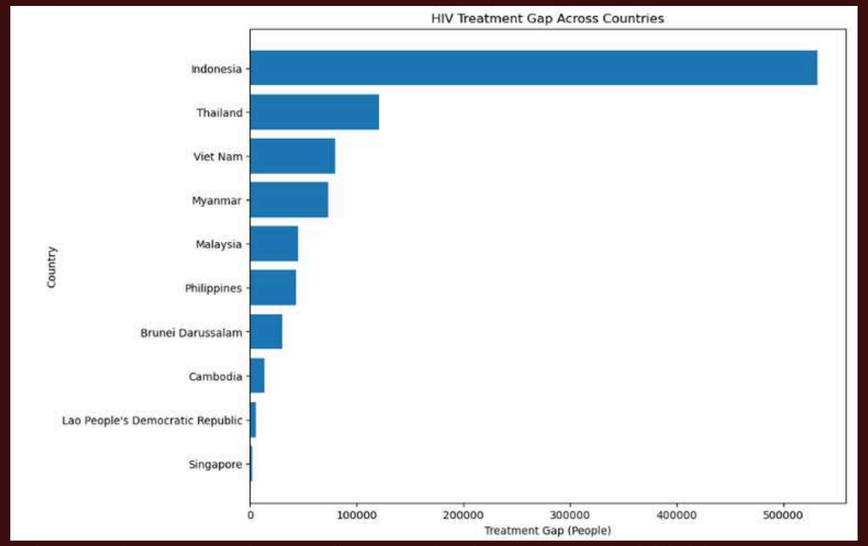
STEP 4 — INEQUALITY ANALYSIS

ART coverage (%)



- Cambodia , Singapore and Thailand have the highest coverage (above 75%) , indicating strong treatment access.
- In contrast ,"Indonesia stands out as an outlier with the lowest ART coverage and highest treatment gap."
- Other countries fall in moderate range (40-70%) .
- Overall , the graph shows that there is clear inequality in ART access across ASEAN countries

Treatment Gap

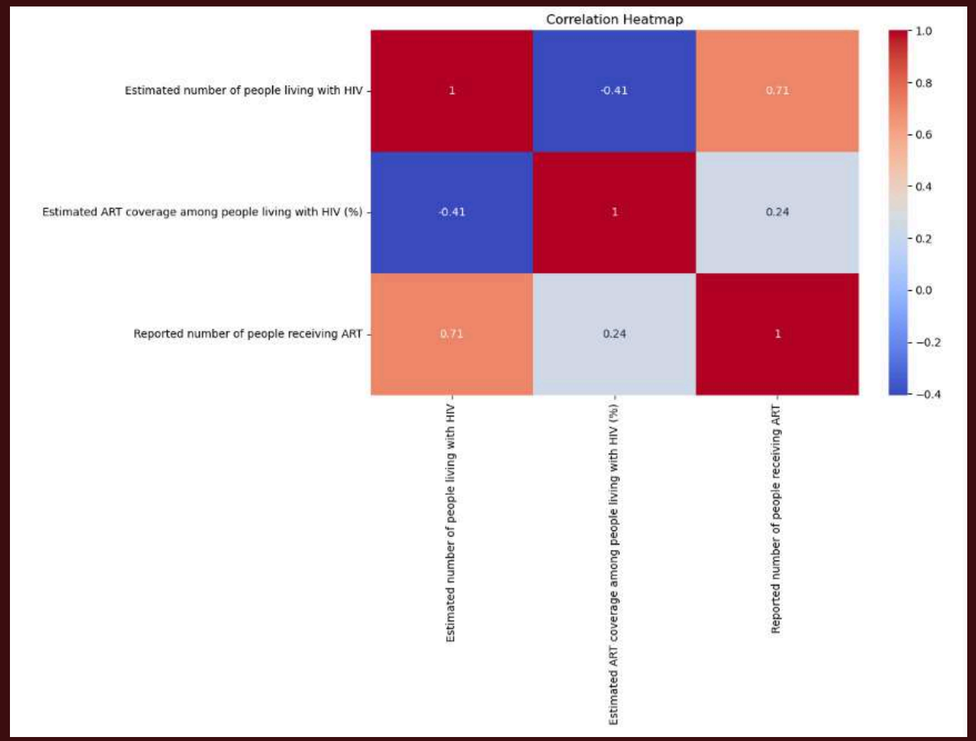


- This graph explain number of people living with HIV but not receiving ART.
- Indonesia has the largest gap by a wide margin , indicating lack of treatment access.
- Countries like Thailand , Viet Nam and Myanmar show moderate gap , while Singapore , Lao PDR and Cambodia , have very small gap , suggest better treatment coverage .
- Overall , this graph highlight clear inequality in access to HIV treatment across countries.

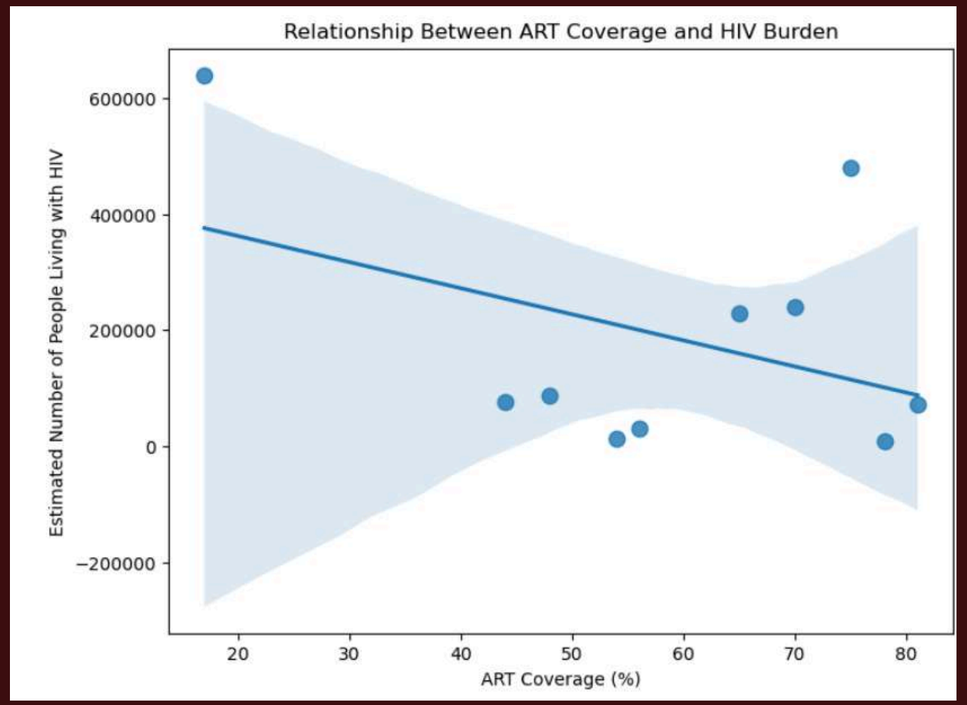
This analysis highlights clear inequality in ART access across countries.
Nations with lower ART coverage , tend to have significantly larger treatment gap , indicate limited healthcare access . This pattern is consistent with the negative relationship observed between ART coverage and HIV burden.

RELATIONSHIP ANALYSIS

Heatmap



Scatter Plot



- The wide spread data points (scatter plot) suggest variability across countries)
- This heatmap and scatter plot indicate a moderate negative correlation between ART coverage and HIV burden (r= -0.41) .
- Increasing ART coverage , helps reduce HIV burden (although other factors also play a role)
- Strong positive correlation (r = 0.71) between HIV burden and number receiving ART
- It reflects that countries with higher HIV cases tend to provide more treatment

DATA ANALYSIS FINDINGS

- ART coverage varies across countries , indicating inequality
- Higher ART coverage → smaller treatment gap
- Indonesia is an outlier with the largest treatment gap
- Moderate negative correlation between ART coverage and HIV burden(r = -0.41)
- Strong positive correlation between HIV burden and number receiving ART (r = 0.71)



DATA-DRIVEN STRATEGIES

Based on the analysis , several data-driven strategies can be proposed by :



- Increase ART accessibility in low-coverage countries to reduce treatment gaps
- Prioritize high-gap countries such as Indonesia for targeted interventions
- Enhance early HIV detection through awareness and testing programs
- Strengthen healthcare systems and funding to improve treatment distribution

ANTIRETROVIRAL THERAPY (ART)

Which drugs are used in antiretroviral therapy?

There are many different types of antiretroviral medications. Each one uses a different strategy, at different points in HIV replication, to stop HIV from making more copies of itself. Types of ART medications include:

- Entry inhibitors.
- Capsid inhibitors.
- Nucleoside reverse transcriptase inhibitors (NRTIs).
- Non-nucleoside reverse transcriptase inhibitors (NNRTIs).
- Integrase inhibitors/integrase strand transfer inhibitors (INSTIs).
- Protease inhibitors.
- Pharmacokinetic enhancers.
- Combination medications.

What is antiretroviral therapy (ART) ?



Antiretroviral therapy (ART) is a combination of medications used to treat HIV (human immunodeficiency virus). It works by stopping HIV from reproducing. It can reduce your levels of HIV and keep your immune system healthy. It’s not a cure, but many people reach undetectable levels of HIV.

BENEFITS OF ANTIRETROVIRAL THERAPY (ART)

- *Keeps your T-cell counts higher and your immune system healthy.*
- *Lowers your risk of having a weakened immune system (immunocompromised) and developing serious or fatal infections.*
- *Lowers the risk you'll develop AIDS.*
- *Lowers your risk of developing certain cancers that are more common among people with HIV (like Kaposi sarcoma, non-Hodgkin lymphoma and others).*
- *Prevents HIV from spreading to others through sex if you have an undetectable viral load.*



4 major routes/causes of transmission HIV

unprotected vaginal or anal sexual intercourse

transmitted from a mother to her infant during pregnancy, childbirth and breastfeeding.

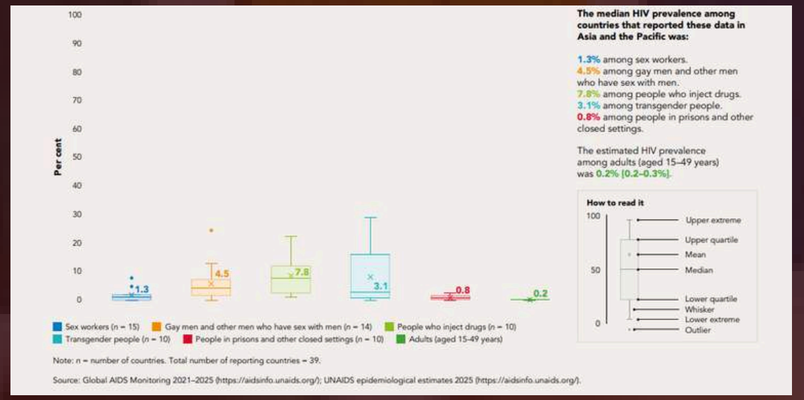
the sharing of contaminated needles, syringes, surgical equipment or other sharp instruments.

transfusion of contaminated blood

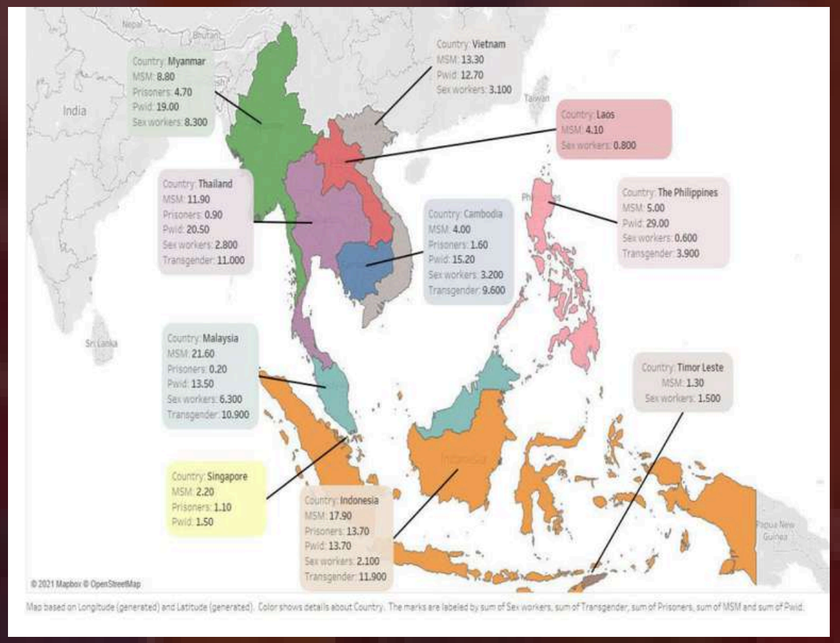
SOURCE : WORLD HEALTH ORGANISATION (WHO)

CAUSES AND TARGET POPULATIONS OF HIV/AIDS

HIV prevalence among people from key populations compared with adults reporting countries in Asia and the Pacific, 2020–2024



SOURCE : UNAIDS 2025 UPDATE



SOURCE : REPORT FROM NATIONAL UNIVERSITY OF SINGAPORE ON UNAIDS 2020 DATA

KEY POPULATION

Transgender

Gay men

Sex workers

People who inject drugs

People in prisons and other closed settings

CRIMES/CASES RELATED TO INFECTION OF HIV/AIDS

CRIME STATISTICS IN MALAYSIA

ACTS INVOLVING CONTROLLED DRUGS OR OTHER PSYCHOACTIVE SUBSTANCES

Number of arrested individuals with positive urine tests 156,803 persons

INJURIOUS ACTS OF A SEXUAL NATURE

Sexual crime 7,740 cases

Number of convicted prisoners admission

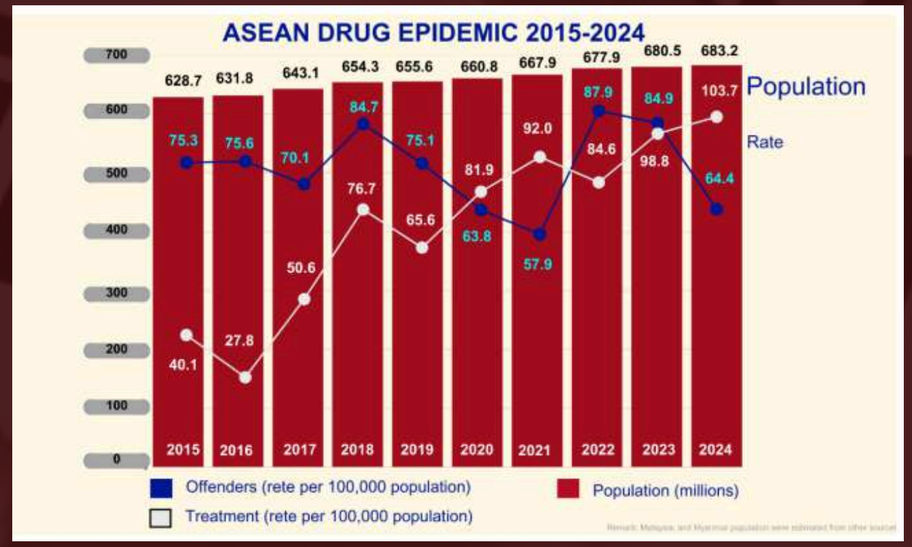
Male 117,211 persons

Female 11,705 persons

Source: Crime Statistics, Malaysia, 2025, Department of Statistics Malaysia (DOSM)

ASEAN CHALLENGES

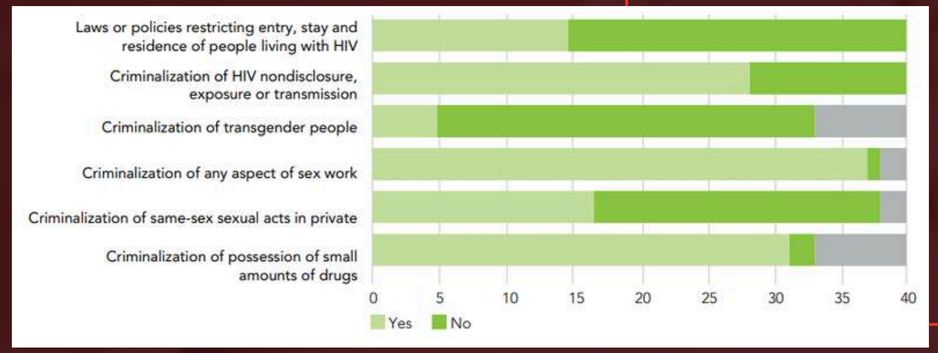
Population and Drug Epidemic in ASEAN



Source: ASEAN Narcotics Cooperation Center Reported on 2025

Number of countries with HIV-related discriminatory and punitive laws, Asia and the Pacific, 2024

SOURCE : UNAIDS 2025 UPDATE





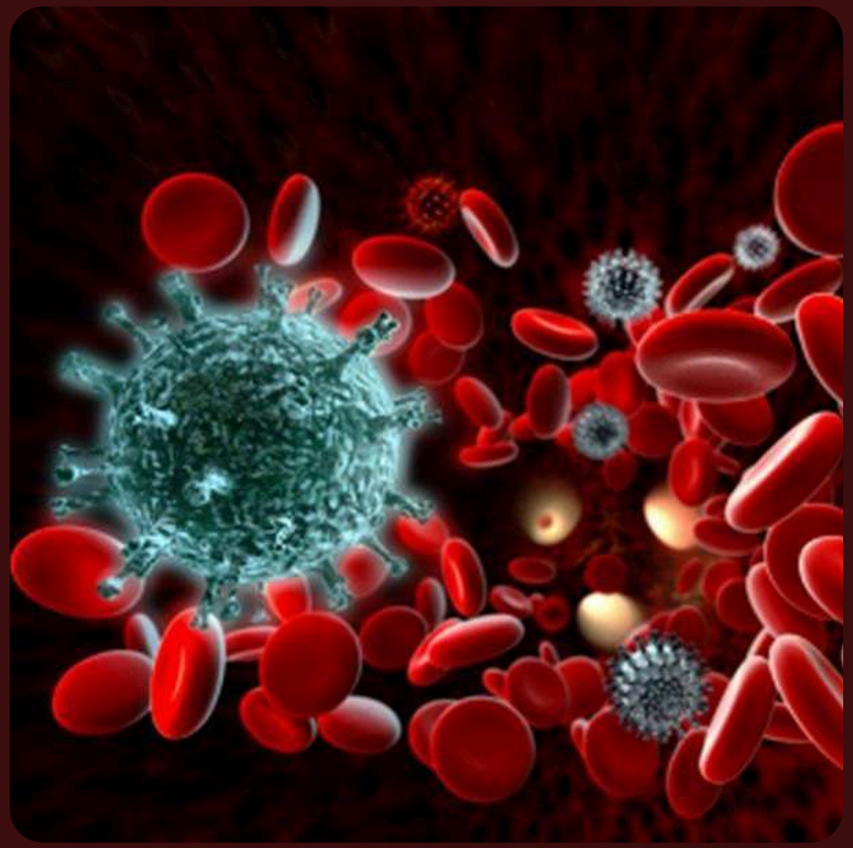
CONCLUSION

In conclusion, the analysis reveals clear inequality in ART access across ASEAN countries. Addressing HIV/AIDS disparities requires not only medical interventions but also inclusive and precise statistical approaches.

Despite notable progress, disparities in healthcare access, data availability, and social acceptance continue to limit effective solutions.

By strengthening data collection systems, ensuring representation of underrepresented groups, and promoting regional cooperation, ASEAN nations can develop more targeted and equitable programs.

Ultimately, data-driven decision-making is essential for reducing inequality and advancing efforts to eliminate HIV/AIDS as a public health threat.



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THANK YOU